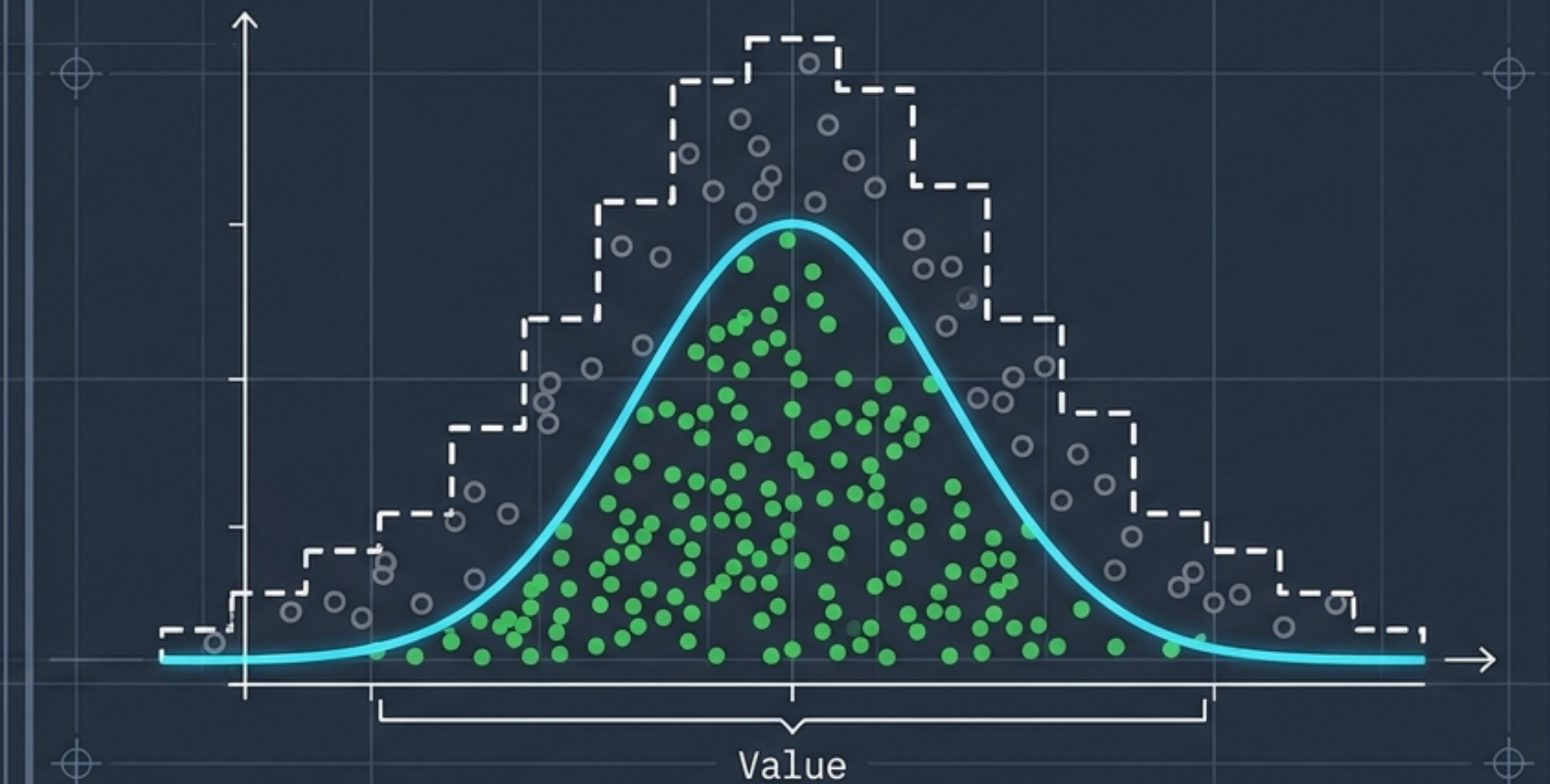


Probability Density



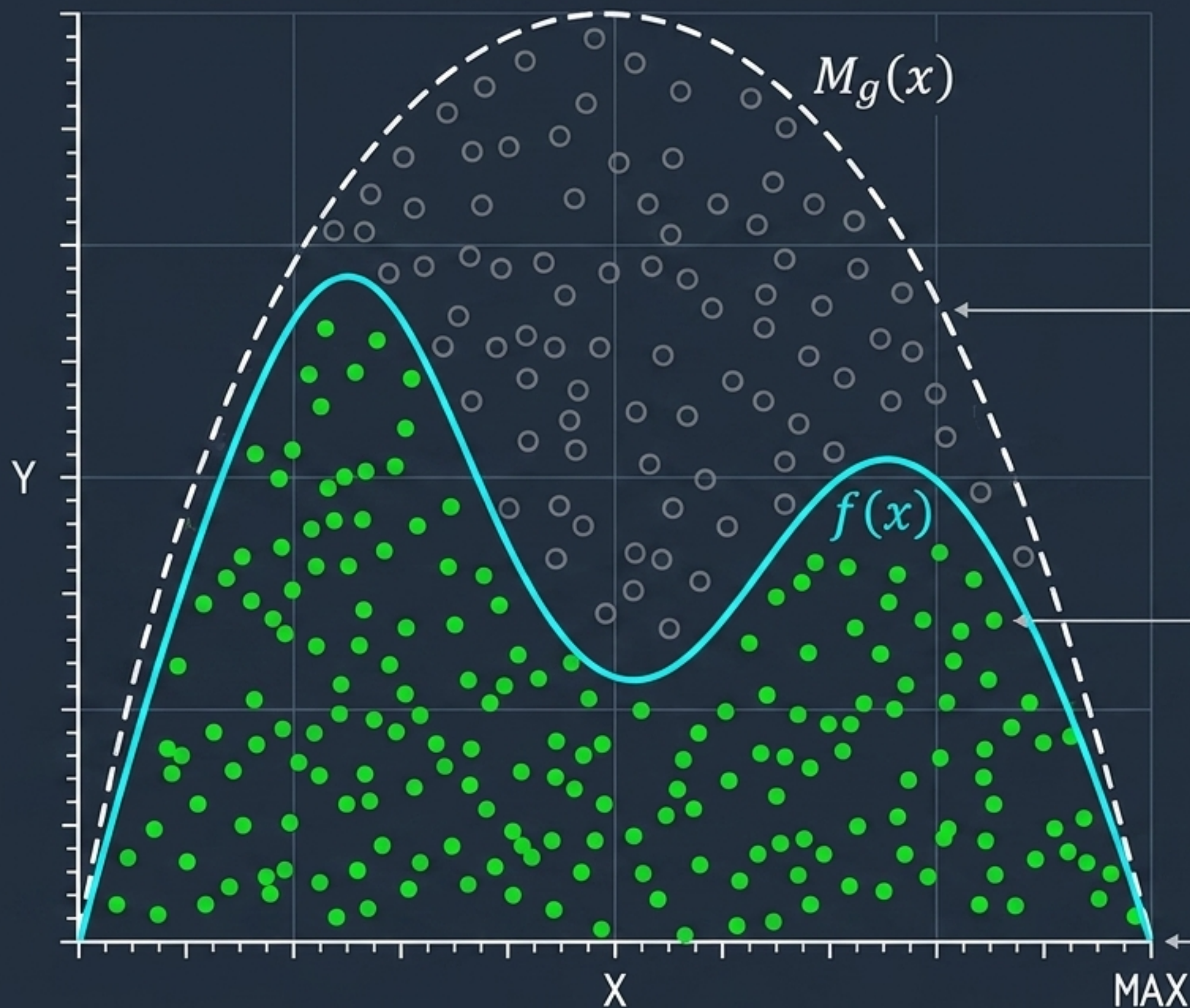
The Acceptance-Rejection Method

Simulating Complex Distributions without the Quantile Function

Lecture 3.4 – Theory, Proofs, and Implementation

Method	Prerequisites	Limitations
Uniform PRNG	Needs only Seed & Modulus.	Only produces Uniform(0,1) numbers. 
Inverse Transform	Needs closed-form Quantile Function (F^{-1}).	Fails on Gamma (non-integer), Standard Normal, and most Bayesian posteriors. 
Acceptance-Rejection 	Needs target density $f(x)$ and a known proposal $g(x)$.	Requires optimizing a bounding constant M .

The closed-form bottleneck: How do we sample from a distribution when we cannot invert its CDF? We change the geometry of the problem.



1. If we scatter points uniformly under an outer proposal envelope...

2. ...the points that happen to fall under the inner target curve are exactly uniformly distributed under that target.

3. The horizontal (x-axis) coordinates of the green points are exact, unapproximated draws from the target distribution.

Target Density $f(x)$

The complex distribution we want to sample from.

We only need the ability to evaluate it for a given x , even if only up to an unknown constant.

Proposal Density $g(x)$

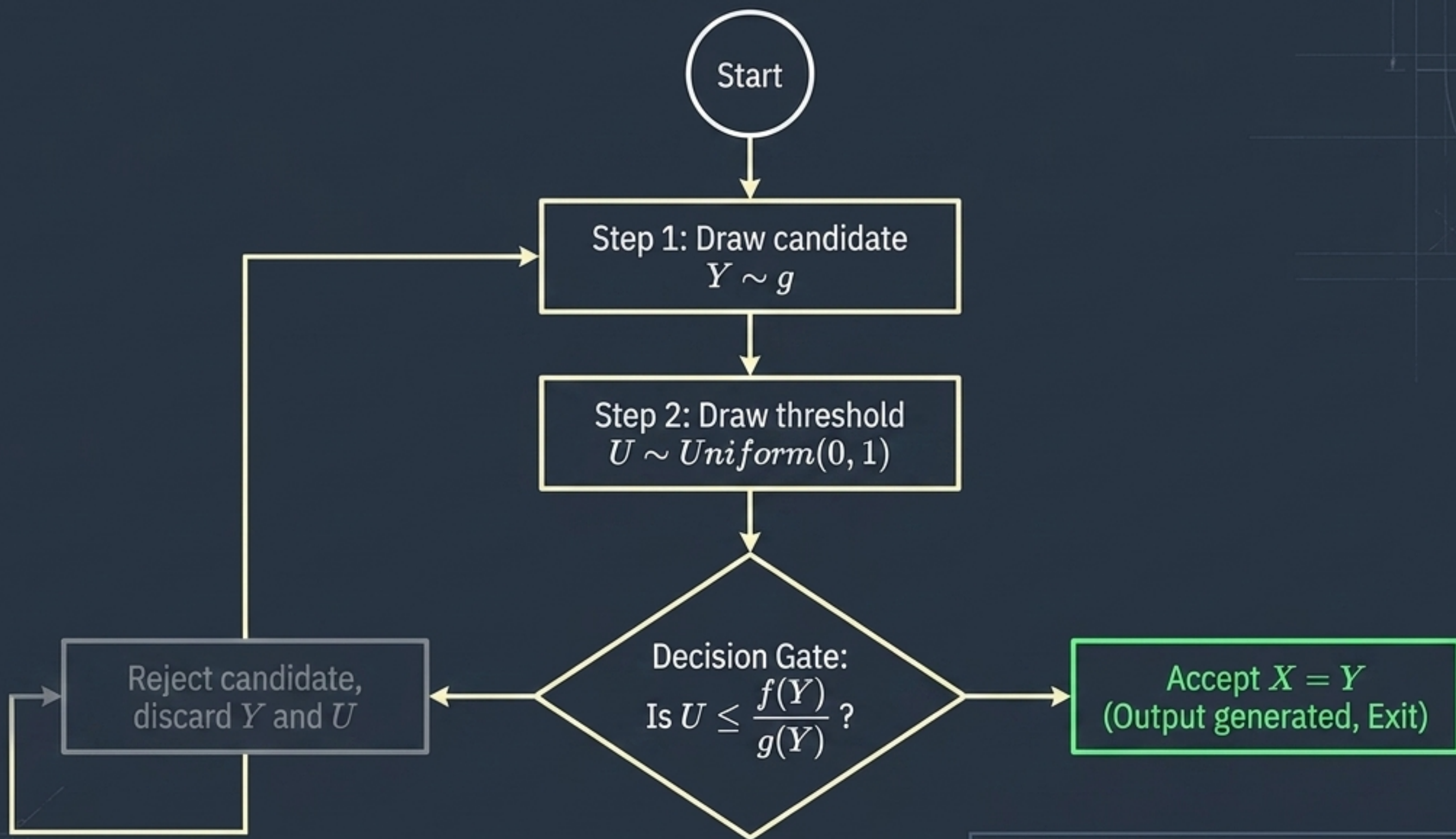
A simpler distribution we already know how to sample from efficiently (e.g., Uniform, Normal).

Bounding Constant M

A scalar $M \geq 1$ guaranteeing the proposal always covers the target.

The Envelope Condition: $f(x) \leq M g(x)$ for all x .

The scaled proposal curve must never dip below the target curve.



Key Insight Panel

Runtime Efficiency Insight: Notice what is missing. This runtime loop requires no integrals and no CDF inversions—only a single numeric comparison.

Theorem 4.1 (Part 1) — The Acceptance Probability

$$P(\text{accept} \mid Y = y) = \frac{f(y)}{Mg(y)} \quad \left] \longrightarrow \right.$$

Conditional on drawing a specific candidate y , the chance we keep it is the ratio of the target height to the envelope height at that exact point.

$$P(\text{accept}) = \int \left[\frac{f(y)}{Mg(y)} \right] g(y) dy \quad \left] \longrightarrow \right.$$

To find the overall success rate, we average this acceptance probability over all possible candidates drawn from $g(y)$.

$$= \frac{1}{M} \int f(y) dy = \frac{1}{M} \quad \left] \longrightarrow \right.$$

Because $f(y)$ is a valid probability density, its total area is exactly 1. Therefore, the global probability of accepting any trial is $1/M$.

Theorem 4.1 (Part 2) — Exact Density Validation

Step 1: Joint probability of drawing a candidate $Y \leq x$ AND accepting it.

$$P(Y \leq x, \text{ accept}) = \int_{-\infty}^x [f(y)/Mg(y)] g(y) dy = F(x)/M$$

Step 2: Apply conditional probability to find the CDF of the accepted value X .

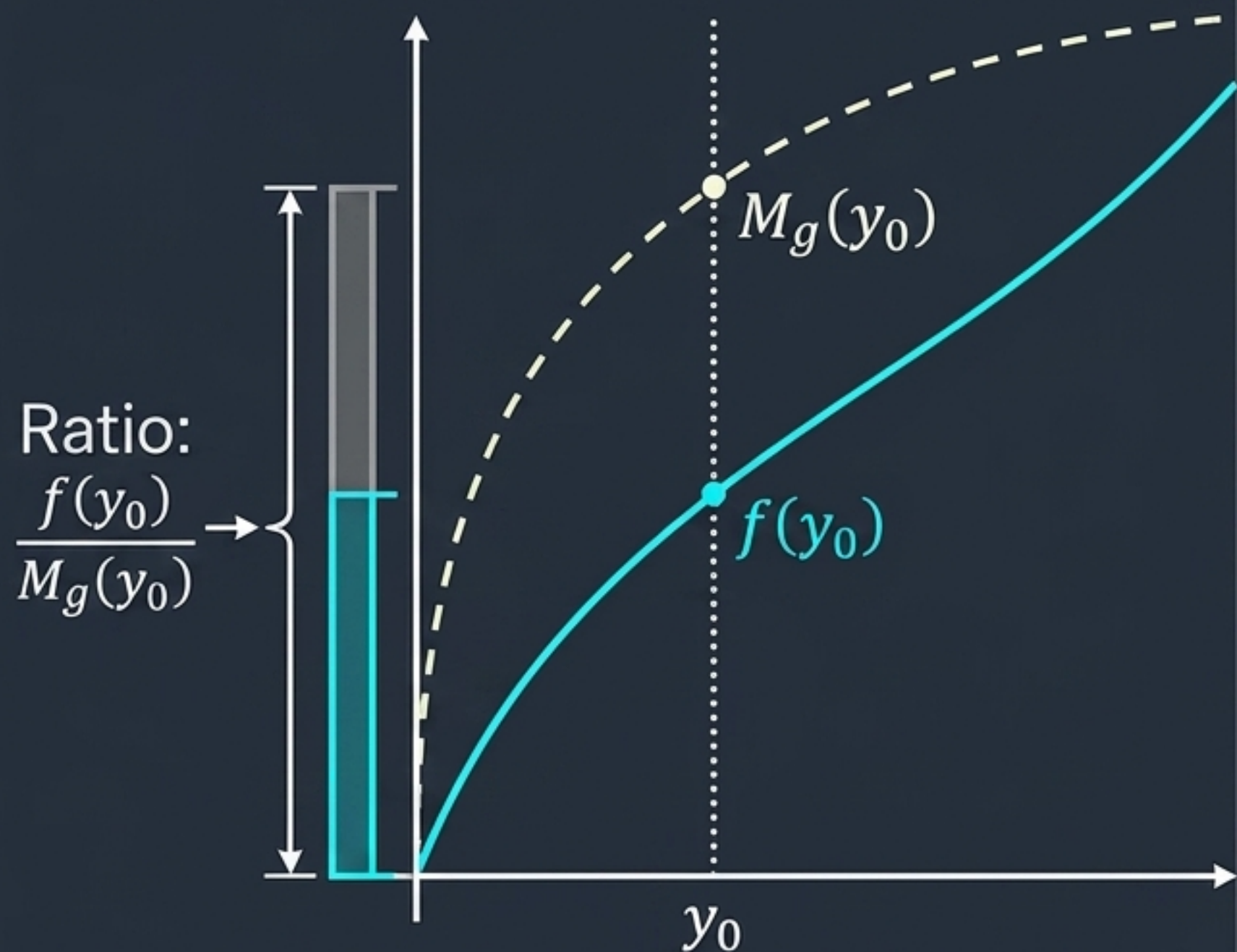
$$P(X \leq x) = P(Y \leq x \mid \text{accept}) = P(Y \leq x, \text{ accept}) / P(\text{accept})$$

Step 3: The Substitution.

$$= [F(x)/M] / [1/M] = F(x)$$

Takeaway: The CDF of our accepted points is exactly $F(x)$. The method is mathematically flawless. There is no approximation, regardless of how loose the envelope is.

The Geometry of the Decision



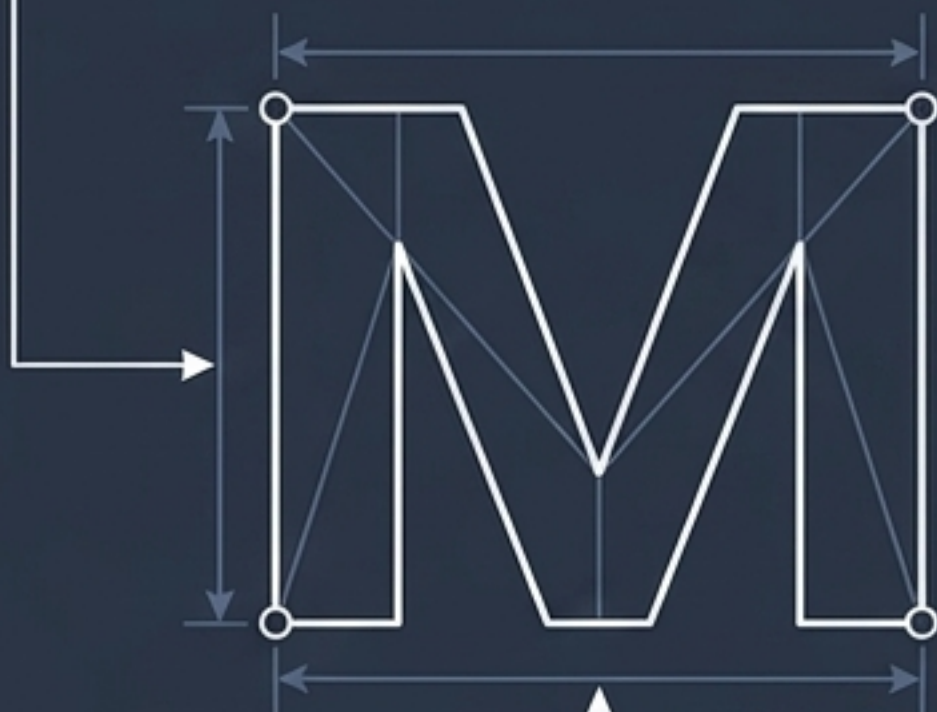
Drawing $U \sim \text{Uniform}(0,1)$ is geometrically equivalent to picking a random point on this single vertical line.

If the random point falls in the lower segment (under the **cyan** target curve f), it is accepted.

If it falls in the upper segment (between f and M_g), it is rejected.

Because each trial succeeds with independent probability $p = 1/M$, the number of trials needed to output one accepted value follows a $\text{Geometric}(1/M)$ distribution.

The Expected Value (mean trials per success) is exactly M .



If $M = 3$, the algorithm generates and discards an average of 3 candidates for every 1 valid draw kept.



The Optimization Goal: **Minimize** M to minimize computational waste.

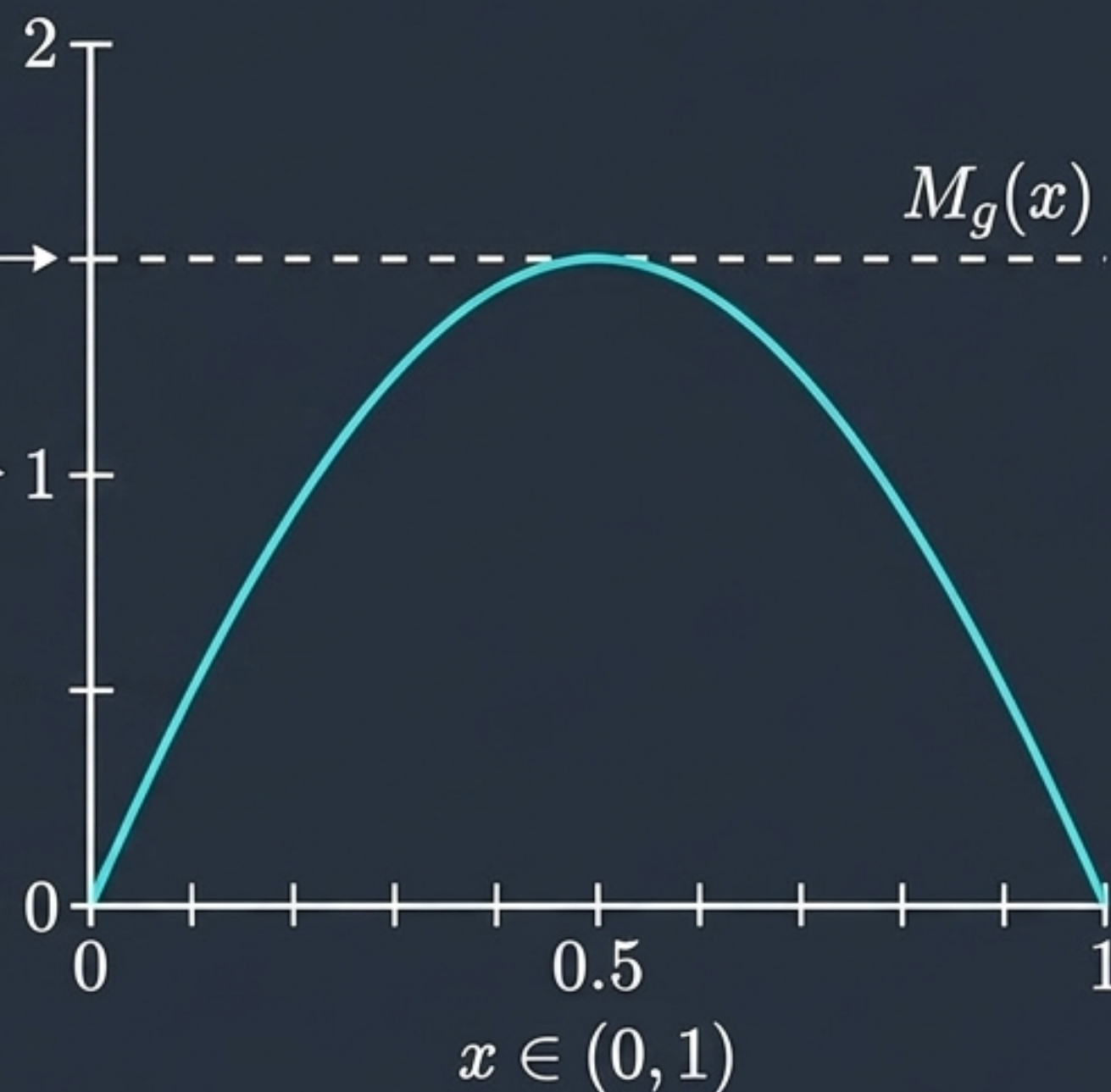
Worked Example: Sampling a Beta(2,2) Distribution

Target: $f(x) = 6x(1-x)$ for $x \in (0, 1)$

Proposal: $g(x) = 1$ (Uniform(0, 1))

Bounding Constant: $M = 1.5$

The Envelope Check: $1.5 \geq 6x(1-x)$
holds true for all $x \in (0, 1)$.



Insight: On average, we will need 1.5 uniform draws per valid Beta draw.

The Tail (High Rejection)

Candidate: $y = 0.9$

Target Height: $f(0.9) = 6(0.9)(0.1) = 0.54$

Threshold Ratio: $0.54 / 1.5 = 0.36$

Random Draw: $U = 0.5$

Result: Since $0.5 > 0.36$,

REJECT

The Peak (High Acceptance)

Candidate: $y = 0.5$

Target Height: $f(0.5) = 6(0.5)(0.5) = 1.5$

Threshold Ratio: $1.5 / 1.5 = 1.0$

Random Draw: $U = 0.8$

Result: Since $0.8 \leq 1.0$,

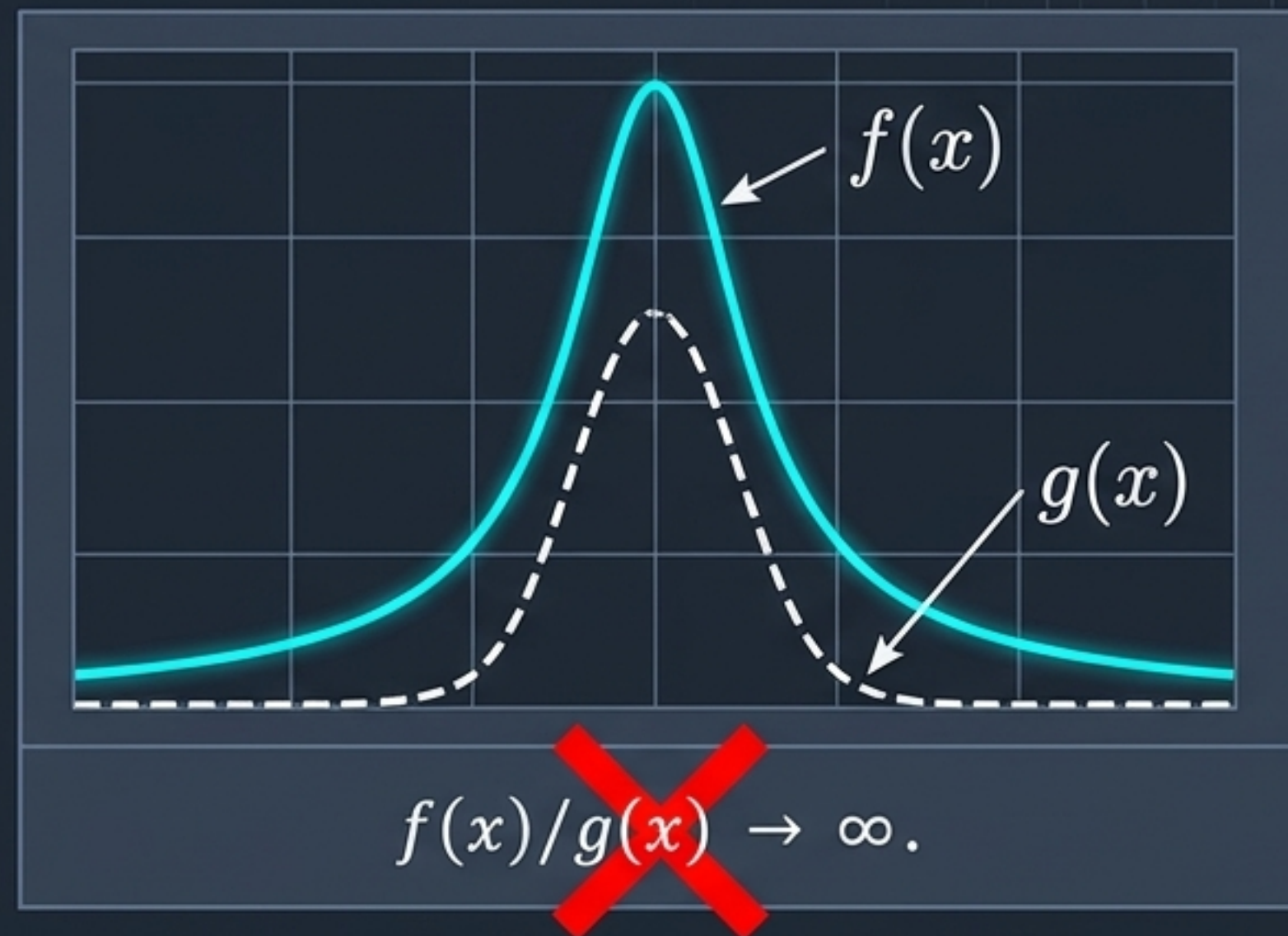
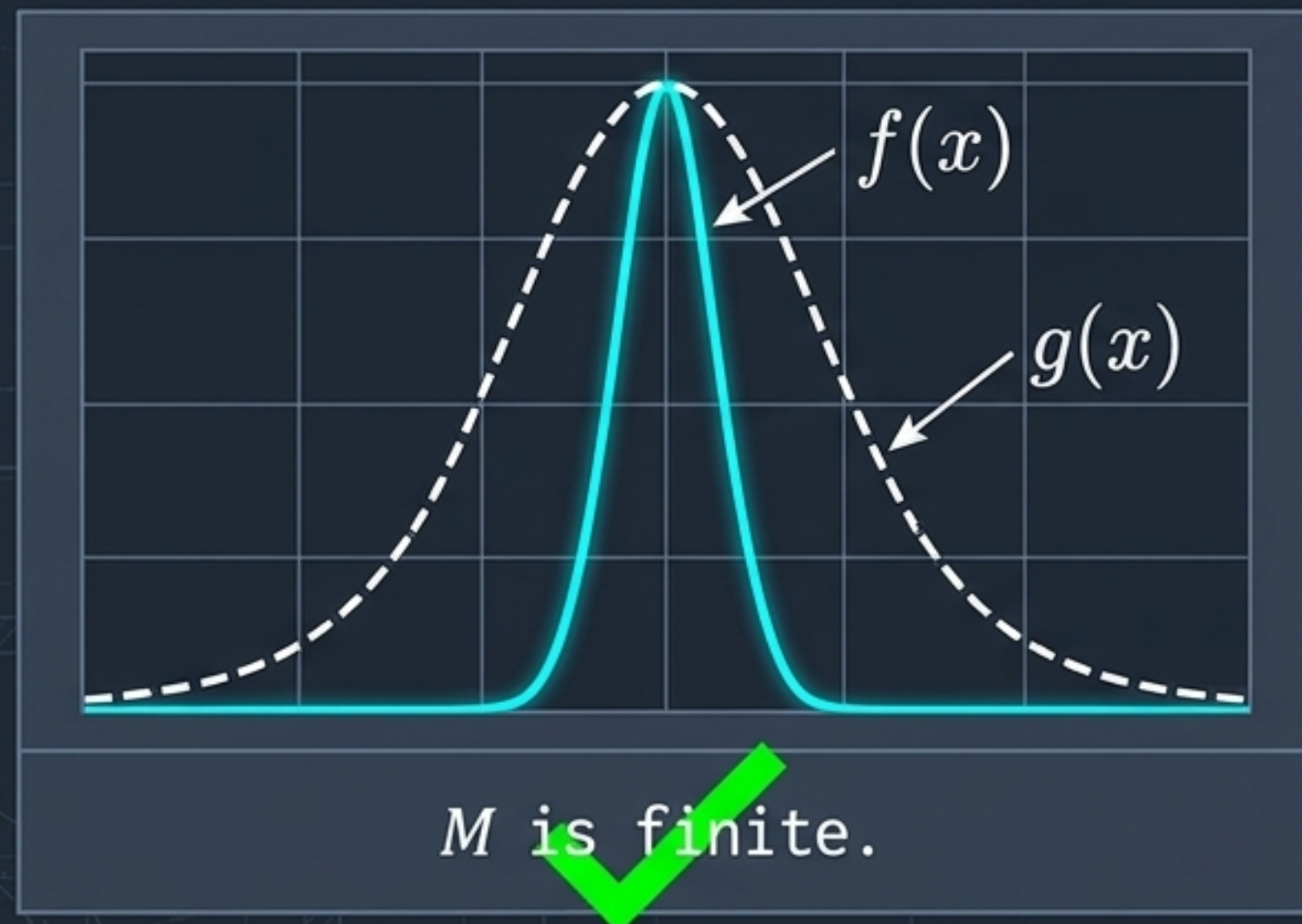
ACCEPT

This differential acceptance rate is exactly what forces the flat uniform proposal into the curved shape of the Beta distribution.

The Optimization Problem: Choosing $g(x)$

The Optimization Rule: You cannot choose just any proposal distribution. The tails dictate the bounds. To achieve a finite, bounded M , the proposal $g(x)$ must have tails at least as generous (heavy) as the target $f(x)$.

For example: A Normal proposal can never bound a Cauchy target, because the Cauchy density decays far slower in the tails.



Precision Blueprint



The Problem:

Sampling rare events (e.g., $Z > 2$). Naive sampling draws from the whole distribution and rejects 98% of points. M becomes massively wasteful.

The Solution: Exponential Tilting.

We build a proposal density that intentionally concentrates its mass exactly where the target region lies.

The Formula:

$$g_{\theta}(x) \propto e^{\theta x} h(x)$$

We tilt a base density h toward the target tail using parameter θ .

Scoreboard

$Z > 2$ from: Standard Normal distribution

Method 1: Naive Rejection

- **Acceptance Probability:** $Q(2) = 0.02275$
- **Cost:** Requires ≈ 44 trials per single success.

Method 2: Exponential Tilting

(Base Exponential shifted to $a=2$, rate $\theta=2$)

- **Bounding Constant:** $M^* = \frac{\phi(2)}{2Q(2)} = 1.187$
- **Acceptance Probability:** $\frac{1}{1.187} = 0.843$
- **Cost:** Requires ≈ 1.2 trials per single success.

The Advantage: The tilted method is ~ 37 times more efficient. Crucially, as we move deeper into the tail ($a \rightarrow \infty$), $M^* \rightarrow 1$. The method becomes more efficient the rarer the event.

Synthesis: The Random Variate Decision Tree

